
CHI-SQUARE TEST

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INTRODUCTION

- Chi-square test is a significant test that provides an alternative technique of determining whether a difference between two proportions is statistically significant.
 - Chi-square test is a non-parametric test that does not rely on any variable distribution or underlying assumptions.
 - Chi-square test was developed by Karl Pearson in 1900.
 - This test measures the difference between what is **observed** and what is **expected** from **categorical** variables following some assumed hypothesis.
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IMPORTANT TERMS

Hypothesis

- **Null Hypothesis (H₀)** The Null Hypothesis is the assumption that the event will not occur. A null hypothesis has no bearing on the study's outcome unless it is rejected. **H₀** is the symbol for it.
 - **Alternate Hypothesis(H₁ or H_a)** - The Alternate Hypothesis is the logical opposite of the null hypothesis. The acceptance of the alternative hypothesis follows the rejection of the null hypothesis. **H₁** is the symbol for it.
 - **Degree of freedom:-** In the Chi-square test Degree of freedom = (No. of rows -1)(No. of columns-1) = $(r-1)(c-1)$.
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IMPORTANT TERMS

- **PARAMETRIC TEST:** The test in which, the population constants like mean, standard deviation, standard error, correlation coefficient, proportion etc. and data tend to follow one assumed or established distribution such as normal, binomial, poisson, etc
 - **NON-PARAMETRIC TEST:** The test in which no constant of a population is used. Data do not follow any specific distribution pattern, and no assumption are made in these tests. For example, to classify good, better and best we just allocate arbitrary numbers or marks to each category.
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IMPORTANT TERMS

CONTINGENCY TABLE: This table is prepared by enumeration of qualitative data by entering the actual frequencies, and if that table represents occurrence of two sets of events, that table is called the contingency table. It is also called as an association table.

APPLICATION OF CHI-SQUARE TEST

Chi-square test is a statistical test used to determine whether there is a significant association between two categorical variables. It compares the observed frequencies in a contingency table to the expected frequencies under the assumption of independence between the variables. The test is widely used in various fields to analyze data and draw conclusions about relationships between categorical variables. Here are some common applications of Chi-square test:

- Educational Testing
 - Medical Research
 - Genetics and Biology
 - Contingency Table Analysis, etc.
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CALCULATION OF CHI-SQUARE TEST

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where, O = Observed Frequency E = Expected Frequency

If two distributions (observed and theoretical) are exactly alike, then $\chi^2 = 0$.

But generally, due to sampling error, χ^2 is not equal to zero.

CALCULATION STEPS

- 1 Calculate the expected frequencies and observed frequencies:
Expected Frequencies (E): The cell frequencies that would be expected in a contingency table if two variables were statistically independent.

Observed Frequencies (O): The cell frequencies actually observed in a contingency table.

$$E = \frac{(\text{column total}) * (\text{row total})}{N}$$

Where N is defined as the Total frequencies.

- 2 Calculation of chi-square test $\chi^2 = \sum \frac{(O - E)^2}{E}$

EXAMPLE

In an Antimalaria campaign in India, Quinine was administered to 500 persons out of a total population of 2000. The no. of fever cases is shown below:

Given Data

Treatment	Fever	No fever	Total
Quinine	20	480	500
No Quinine	100	1400	1500
Total	120	1880	2000

Discuss the usefulness of Quinine in checking malaria:

Here RT defines as Row total, CT is defined as column total and N=Total frequencies.

So Expected value(E)= (RT*CT)/N.

Then $E_{11}=30$, $E_{12}= 470$, $E_{21}=90$, $E_{22}=1410$.

Calculation of Chi-square

O	E	(O-E)	(O-E) ²	(O-E) ² /E
20	30	-10	100	3.33
100	90	-10	100	1.11
480	470	10	100	0.21
1400	1410	-10	100	0.07
				$\chi^2=4.72$

EXAMPLE

So chi-square test $\chi^2 = \sum \frac{(O - E)^2}{E}$
= 4.72.

Degree of freedom = $(c-1)*(r-1)$
= $(2-1)*(2-1)=1$.

Let us take the level of significance be 5 percent i.e 0.05.

From Chi-square table we get tabulated ($\chi_{0.05}^2$) value be 3.84.

Here Expected chi-square value be greater then tabulated Chi-square value. So null hypothesis be failed and rejected.

Hence Quinine is useful in checking of Malaria.

A **chi-square value** table lists critical values for a chi-square test, which are determined by the test's degrees of freedom (df) and the significance level (alpha). To use the table, find the row corresponding to your degrees of freedom and the column corresponding to your significance level, then read the chi-square value at their intersection. These values serve as thresholds to determine if the results of a chi-square test are statistically significant.

CONDITIONS

► Steps :

- Create a contingency table with observed frequencies.
- Calculate the expected frequencies using :

$$E_{ij} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

- Compute Chi-Square :

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- If $\chi^2(\text{Calculated}) > \chi^2(\text{tabulated})$, reject the Null hypothesis. Otherwise, we do not reject the null hypothesis i.e The data does not show a significant difference.

CONDITIONS FOR CHI-SQUARE TEST

The following conditions should be satisfied before the chi-square test can be applied.

- The data must be in the form of frequencies.
- The frequency data must have a precise numerical value and must be organized into categories or groups.
- observations recorded and used are collected on a random basis.
- All the items in the sample must be independent.
- No group should have only a few things, say fewer than 10, and the total number of items must be at least 50.

LIMITATIONS

- The chi-square test does not give us much information about the strength of the relationship. It only conveys the existence or non-existence of the relationships between the variables investigated.
 - The chi-square test is sensitive to sample size. This may make a weak relationship statistically significant if the sample is large enough. Therefore, chi-square should be used together with appropriate measures of association (lambda, Cramer's V or gamma) to guide in deciding whether a relationship is important and worth pursuing.
 - The chi-square test is also sensitive to small expected frequencies. It can be used only when not more than 20 percent of the cells have an expected frequency of less than 5.
 - Cannot be used when samples are related or matched.
 - Affected by grouping. The way data is grouped or categorized can affect the outcome of the test.
 - Can work effectively with categorical data but not suitable for numeric or continuous data.
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